How can slip stick friction be maximized between finger and computer mice

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1 Introduction

Friction is a topic that has intrigued people for millenia, and pioneered to this very day. Think honestly, what causes friction? Small microscopic asperities? How does perfectly smooth rubber still present friction? Or jagged ice is so slippery? Is water the issue? Then why can you tread on water? From scientists, control engineers, and frankly, scholars that experienced its' effects in other forms, all trying to understand this confounding concept. Namely, slip-stick friction is the limit cycle of an objects' continuous oscillation between slipping and sticking, such as seen in violins, door's creakings, badly engineered anything. It is a topic fixated by engineers in trying to reduce, common solutions include adding grease, oil, etc. However, in the realm of competitive Minecraft, there exists many instances where the desirable trait of clicking fast can be accompanied by exploitation of slip stick atop the mouse. This is referred to in the community as "drag clicking".

In this paper, we will attempt to find ways to test the slip stick properties pertaining to drag clicking by modeling material properties of common methods using advanced friction models such as LuGre and Dahl friction models.

2 Background

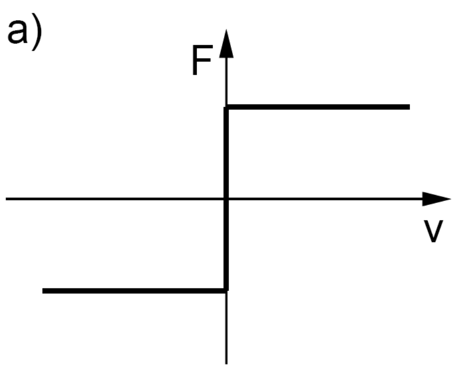

Figure 1: The change of friction with respect to velocity

Our commonly conceived model of friction—the coulomb friction model—has several limitations. It suggests that upon motion the frictional force is linearly proportional to the normal force and of the opposite direction of the velocity [3]. illustrated in figure 1. This is perceived as due to microscopic asperities interlocking with one another. This, for instance explains the independence of apparent surface area and dependence on normal force (i.e. the normal force pushes more asperities in contact, despite small appearant contact area, true microscopic contact area is much larger due to the increased force) However, friction still has much greater depth covering phenomena such as:

1. The Stribeck Effect
2. The Microscale pre-slip deformations
3. Dependence on past events
4. Ultimate Friction aging

2.1 Friction Phenomena

2.1.1 Viscous Friction

Viscous friction introduces the role of velocity in that sliding friction increases with velocity. Though exact instantaneous velocity cannot be provided via discrete data points, a treadmill setup could mitigate this with a constant velocity upon contact

2.1.2 Static Friction

Friction exhibits a noticable decrease when set in motion. Increasing applied force beyond this is usually the exciter for slip-stick friction.

2.1.3 The Stribeck Effect

The Stribeck Effect suggests that static friction does not suddenly jump to viscous friction but happens gradually. The rate of friction mode change depends much on material properties and in particular, we expect sharper jumps to exhibit superior drag-clicking capabilities

2.1.4 Rising Static friction

Refers to the increase in static friction as a result of time. Described as the normal forces gradually increasing true surface area between microscopic asperities over time.

2.2 State Space Equations

To coherently account for behaviour involving state changes, we must shift to dynamic models that uses state space equations. Most namely, the Dahl Model and LuGre Model In state space equations, there exists ”state variables” that capture internal states of the system, when there is more than one, it is packaged into a vector matrix. State space equations define the derivative of state variable as a function of state variables, this gives us the rate of change at an instant, as we integrate it over small discrete units of time it gives us a prediction of the current state (current position, force, etc.). The predicted state variables are fed back into the derivative as linear equation constants to be then integrated again.[4] However, looking at the relationship between variables in a state space equation can describe the stability of a system, which is particularly useful in stick–slip motion.

2.3 The Dahl Model

While the standard textbook definition for friction stem from rigid body microscopic asperities, the Dahl Model instead visualizes two bodies making contact from elastic bristles scattered over the contacting surface[1]. Illustrated in figure 2.

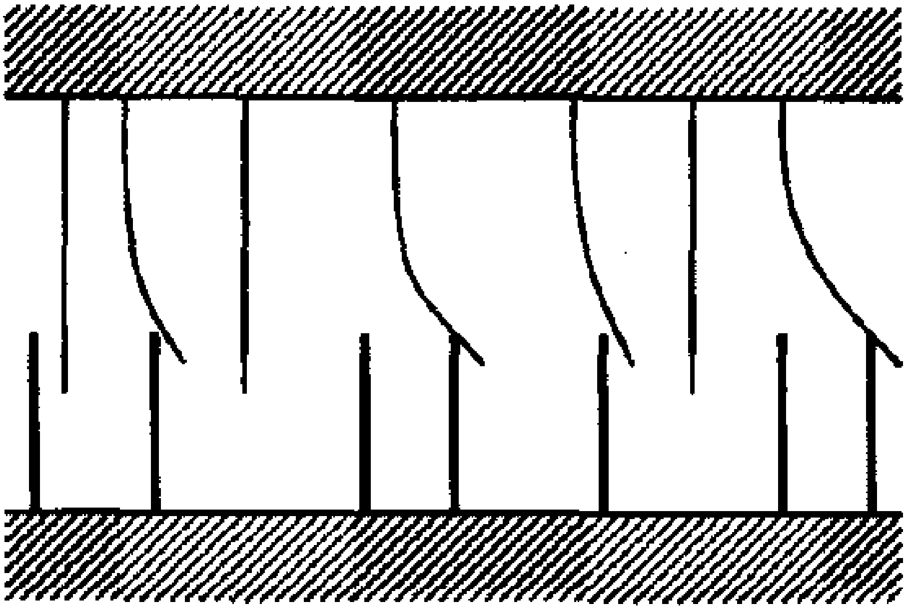

Figure 2: Depiction of Dahl Model's Bristles [1]

The Dahl model is defined as the following:

$$F = F_c \left(1 - e^{-\frac{\sigma_0 |x|}{F_c}}\right) \operatorname{sgn}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) \tag{1}$$

Where F_c is the coulomb friction of the surface, x is the displacement from original position (this is primarily focused on the microscopic displacements), and σ_0 is the "stiffness" of microscopic bumps and affects the shift to maximum coulomb friction.

To use this for state space equations we must find the derivative using chain rule:

$$\frac{\mathrm{d}F}{\mathrm{d}x} = \cancel{-}\cancel{F_c} \cdot e^{-\frac{\sigma_0 |x|}{F_c}} \cdot \cancel{-}\frac{\sigma_0}{\cancel{F_c}} \operatorname{sgn}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) \operatorname{sgn}(x) \tag{2}$$

$$= \sigma_0 e^{-\frac{\sigma_0 |x|}{F_c}} \operatorname{sgn}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) \operatorname{sgn}(x) \tag{3}$$

and to put the actual current force state back into this derivative function (since state space equations depend on previous states), we rearrange eq (1) to isolate $e^{-\frac{\sigma_0 |x|}{F_c}}$ like such:

$$F = F_c \left(1 - e^{-\frac{\sigma_0 |x|}{F_c}}\right) \operatorname{sgn}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) \tag{4}$$

$$\frac{F}{F_c} \operatorname{sgn}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) = 1 - e^{-\frac{\sigma_0 |x|}{F_c}} \tag{5}$$

$$e^{-\frac{\sigma_0 |x|}{F_c}} = 1 - \frac{F}{F_c} \operatorname{sgn}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right) \tag{6}$$

and putting it back in the differential equation we get:

$$\frac{\mathrm{d}F}{\mathrm{d}x} = \sigma_0 \left(1 - \frac{F}{F_c}\right) \cancel{\mathrm{sgn}\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2} \mathrm{sgn}\,(x) \tag{7}$$
$$= \sigma_0 \left(1 - \frac{F}{F_c}\right) \mathrm{sgn}\,(x) \tag{8}$$

The Dahl model's equation shares similar properties of an exponential function take of the stress-strain curve (depicted in figure 3), most notably, the point at which it shifts from an elastic spring to continual bristle movement

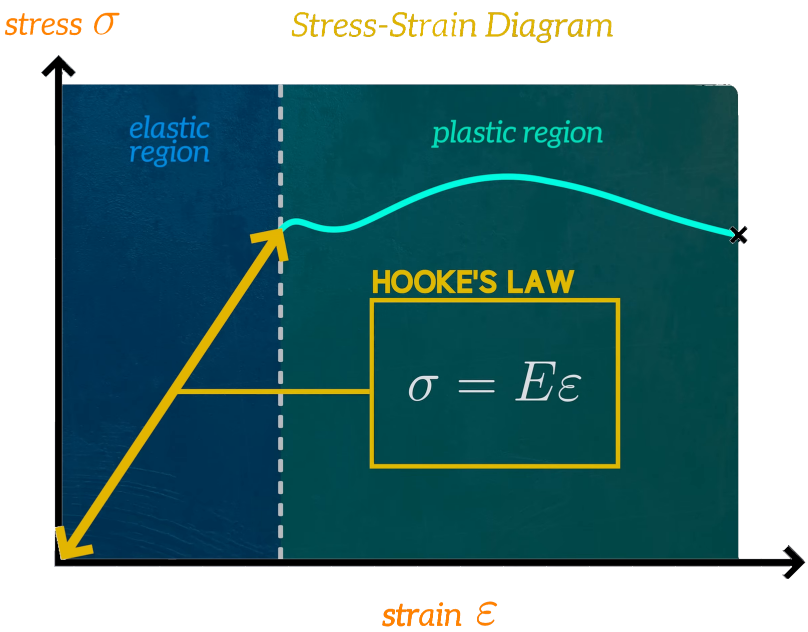

Figure 3: An annotation of a generic stress-strain curve[2]

Most importantly the Dahl Model introduces $z = \frac{F}{\sigma_0}$ as a state variable that represents the "average bristle deflection" [46]. This is very similar to Hooke's law. rearranging, we get:

$$F = \sigma_0 z \tag{9}$$

similarly, in Hooke's law presented in the Physics formula booklet:

$$F = kx \tag{10}$$

Differentiating $z = \frac{1}{\sigma_0} \cdot F$ to find the state equation:

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{1}{\sigma_0} \cdot \frac{\mathrm{d}F}{\mathrm{d}x} \cdot \frac{\mathrm{d}x}{\mathrm{d}t} \tag{11}$$
$$= \frac{1}{\sigma_0} \cdot \frac{\mathrm{d}F}{\mathrm{d}x} \cdot v \tag{12}$$
$$\tag{13}$$

and with $\frac{\mathrm{d}F}{\mathrm{d}x} = \sigma_0 \left(1 - \frac{F}{F_c} \mathrm{sgn}\,(v)\right)$

$$= \frac{1}{\cancel{\sigma_0}} \cdot \cancel{\sigma_0} \left(1 - \frac{F}{F_c} \operatorname{sgn}(v)\right) v \tag{14}$$

$$= \left(1 - \frac{F}{F_c} \operatorname{sgn}(v)\right) v \tag{15}$$

$$\tag{16}$$

now with $F = \sigma_0 z$ and $sgn(v)v = |v|$ we arrive at the definitive Dahl's model

$$= \left(1 - \frac{\sigma_0 z}{F_c} \operatorname{sgn}(v)\right) v \tag{17}$$

$$\frac{\mathrm{d}z}{\mathrm{d}t} = v - \frac{\sigma_0 z}{F_c} |v| \tag{18}$$

[3]

2.4 Lugre Model

The Lugre model takes in consideration that coulomb friction is not constant but rather dependent of velocity, thereby replacing it with $g(v) = F_c + (F_s - F_c)\, e^{-|\frac{v}{v_s}|^{\alpha}}$ that considers the rate of change from F_s to F_c to be non-instantaneous as shown in figure 4

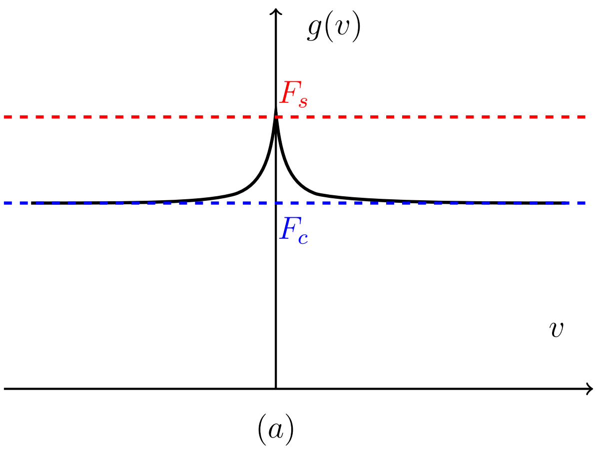

Figure 4: Graph of $g(v)$ with relative F_s and F_c coordinates

to create the following differential equation:

(19)

2.5 Drag clicking specific methods

There exists several known methods in the community to increase the accessibility of drag clicking on a mouse. With any youtube search you will find it populated with applying "tape" to your mouse, along with sharpie markers applied to that, then the mention of keeping your hands dry. In private I have discovered several new methods to increase friction, along with additive methods, I have discovered subtractive methods. Additive methods include rubbing crushed cello rosin dust onto the surface of the mouse with your finger, it was meant toadjunct slip stick properties of stringed instruments, though the effect is universal. Subtractive include rubbing the surface of your mouse with the bottom of your mousepad firmly, or with the leaf of a well hydrated plant. We will measure the rate at which static friction equalizes to coulomb friction with respect to time

3 Method

The constants in the lugre friction model (σ_0, σ_1, etc), will be found by measuring steady friction force at various velocities [3]. This will be done with a calibrated force meter

References

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[2] The Efficient Engineer. *An Introduction to Stress and Strain*. 2020. URL: https://www.youtube.com/watch?v=aQf6Q8t1FQE.

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[4] MATLAB. *Introduction to State-Space Equations — State Space, Part 1*. 2019. URL: https://www.youtube.com/watch?v=hpeKrMG-WP0.